

**LOCAL BUCKLING AND CRIPPLING OF
BEAMS/STRINGERS**

Part: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3130**

Title: **LOCAL BUCKLING AND CRIPPLING OF
BEAMS/STRINGERS**

Revision: **1.1**

Date: **29/09/2021**

1 INTRODUCTION

1.1 OBJECT

This chapter is concerned with the local buckling stress of metallic thin walled sections when loaded in compression, as well as with the ultimate load the section (in a short length) is capable to sustain before failure beyond local buckling, i.e. with the crippling strength.

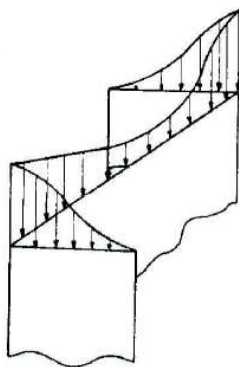


Figure 1. Typical compression stress distribution in a thin-walled profile after local buckling

For sustaining compression loads metals are formed, extruded or machined in order to get composite shapes as angles, channels, zeos, ... with significantly higher buckling stresses than simple flat sheets. Calculation methods for predicting both the local buckling stress and the ultimate compression strength (crippling) of the section normally decompose the section into flat elements, supported on one or two sides, before using available theories for evaluating buckling and crippling stresses in flat plates.

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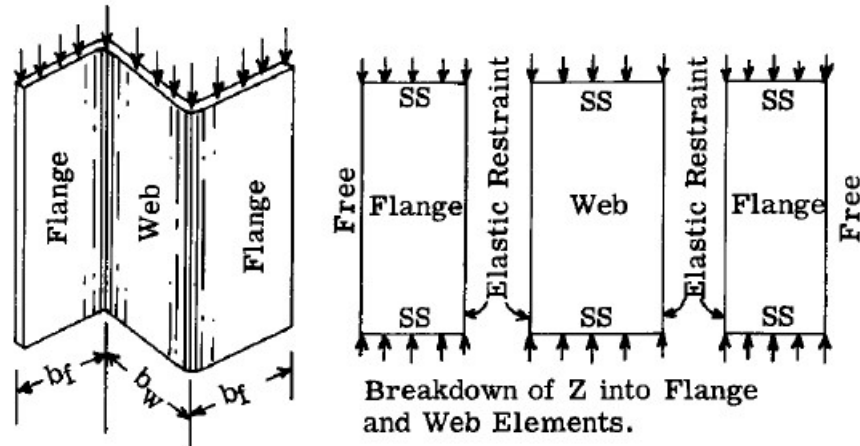


Figure 2. Typical decomposition of a thin walled section into flat elements (Bruhn C6.3)

A section element with one of its sides supported and the other free is typically referred as a flange (or flange element). In Z, J or I shaped sections, the centre elements supported on both sides is typically referred as the web (or web element).

1.2 SCOPE AND LIMITATIONS

Calculation of the crippling strength of thin walled metallic sections. The method has the following limitations:

- ✓ Metallic profiles with thin walled sections
- ✓ Short column range (column buckling not covered herein)

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2 METHOD DESCRIPTION: LOCAL BUCKLING

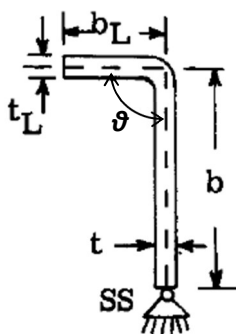
As shall be discussed in following chapter, compression members are capable to sustain load beyond the local buckling onset (of one or more section elements). However, the calculation of the local buckling stress of a section may be of interest for several reasons, including:

- ✓ Variable compression stress across element width (crippling methods only deal with uniform stresses and may be incapable to estimate allowable loads with shear/bending interaction)
- ✓ Setting a cut-off value for allowable stresses in certain types of sections acting as column members with risk of global buckling (e.g. L or T stringers of a stiffened panels)
- ✓ Etc...

The methodology to be used to evaluate the buckling stresses of metallic profiles, after the decomposition of the section into long flat elements (with none or one sides free) shall be that described in HSB 45111-01. For section elements of short width compared to its thickness, the inelastic effects shall be accounted using plastic corrections proposed in ESDU 83044, ESDU 76016 or that in Niu's book (Ref. 7) chapter 11.2(B). Support condition on supported edges shall be simple supports (clamping edges or introducing rotation restraints in calculations shall not be permitted without further explicit validations).

Section elements will be effectively supported on one of its sides by other element/flange only if this supporting (perpendicular) flange is stiff enough. The check may be done using *Windenburg's* methodology (see Bruhn C7.9). If t is the thickness of the (first) element, b its equivalent width (using midline crossings as end points), A_L and I_L the area and moment of inertia of the lateral flange intending to provide support, this support shall be effective if:

$$2.73 \cdot I_L / (b \cdot t^3) - A_L / (b \cdot t) \geq 5$$



Area and inertia of flange intending to provide effective support:

$$A_L = b_L \cdot t_L$$

$$I_L = (t_L \cdot b_L^3) / 12 + A_L \cdot (b_L/2)^2 = (t_L \cdot b_L^3) / 3$$

Figure 3. Parameters in Windenburg check for effective lateral support

In case the angle between the segment and the flange providing lateral support ϑ is not 90° , the width of the supporting flange in above formulas b_L shall be replaced by $b_L \cdot \cos(\vartheta - 90^\circ)$. For angles between segments very acute ($\vartheta < 45^\circ$) or obtuse ($\vartheta > 135^\circ$) this correction might be questionable.

The utility of the *Windenburg* check can be extended to calculate, in case the support is not effective, the reduced width of the section element that the flange could effectively support. This effectively supported width shall be an useful tool later to get the buckling and crippling strengths of non-properly supported elements:

$$b_{\text{eff-sup}} = [2.73 \cdot I_L / t^3 - A_L / t] / 5$$

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3 METHOD DESCRIPTION: CRIPPLING

Tests of short lengths of sections composed of flange-plate elements often shown that after the section has buckled locally, the unit still has the ability to carry a greater load before failure occurs. In other words, the local buckling and local failure loads are not the same.

As the load of the section is increased, the buckles on the flat portions get larger but most of the increasing load is transferred to the much stiffer corner regions until the stress intensity reaches a high enough value to cause sufficient deformation to cause failure (see Figure 1). Previous paragraphs are extracts from Bruhn §C.7.

That ultimate failure in compression beyond local buckling is called crippling. The methods of solution for the crippling ultimate strength are semi-empirical in character, and the results of such methods have been sufficiently proven by tests. The short column is idealized as a number of flat plate elements of low thickness, supported along their sides. The total crippling strength of the section is obtained as the sum of the crippling strengths of all those plate elements.

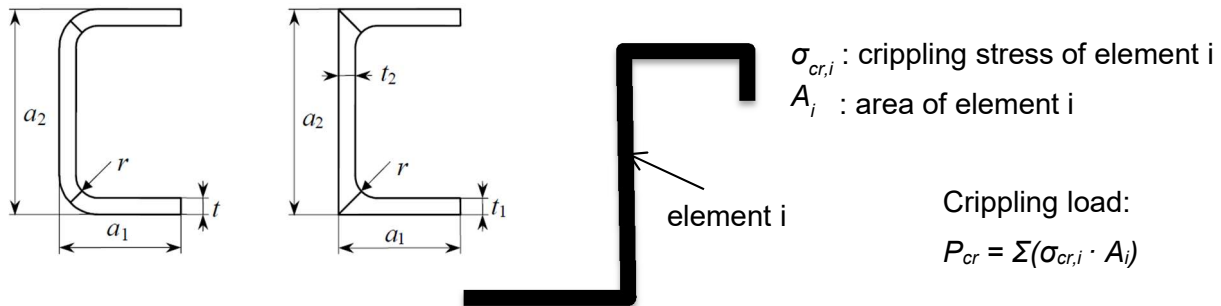


Figure 4. Crippling load as sum of the crippling loads of the section elements/flanges

The applicable method to obtain the crippling strength of a metallic section from its material and geometry data is that described in detail in HSB 53211-01 Issue I (see remarks (*) and (**) below).

Cut off with the yield compression stress is generally recommended for the crippling strength of the complete section.

(*) It has been observed that formulas presented in the HSB for obtaining the crippling width (b) intend to approximate the crippling area of the section element (i.e. the product $b \cdot t$), and that the resulting b dimension may exceed the most outer width of the section element (a). This expression shall be respected to get the crippling area, but the value of b shall be truncated to a maximum of $a \cdot t/2$ when using the HSB expression for the crippling stress.

(**) It has also been observed that, up to some levels of the b/t ratio, the local buckling stress of a section element, assuming the supported lateral sides are simple-supports (rotation free), can be greater than the crippling stress calculated with this methodology. If that occurs, the crippling stress of the section element will be finally made equal to the maximum of those two values.

Lateral support conditions

The crippling stress of each of the section elements is function of the material, the geometry (mainly width and thickness), but also of its support conditions. It depends on having only one or both lateral

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sides “connected” to another adjacent element or not, and also if these (normal) adjacent elements provide sufficient support. Thus, the support conditions of each of the 2 sides of an element can be:

- ✓ **Supported**
- ✓ **Free**

If the edge is not really flat but ends in a lip or a bulb, the lip/bulb area shall be accounted to determine the final crippling strength, and even may change the support condition of this side to supported

Effective support

- ✓ **Elements supported on both sides**

The reference HSB norm defines a procedure to check if lateral flanges provide sufficient lateral support to a section element supported on both sides. Effective lateral support is provided if...

$$a' - t > 0.2 \cdot a$$

...where a , t are respectively the width and thickness of the element being calculated and a' the width of the adjacent flange that should provide the lateral support.

If this check is not true, the support condition of that lateral side shall be set to free, and crippling stress of the element calculated accordingly.

For consistency with the local buckling analysis, the *Windenburg* check presented in previous sub-chapter (see §2, and Figure 3) shall be used instead. It is noticed that the effective-support-condition is reached on similar support widths, with both methods, if thickness of section elements are similar (ratio 1.0 on graph below), and that the *Windenburg* check has the advantage of being sensible to thickness variations between elements.

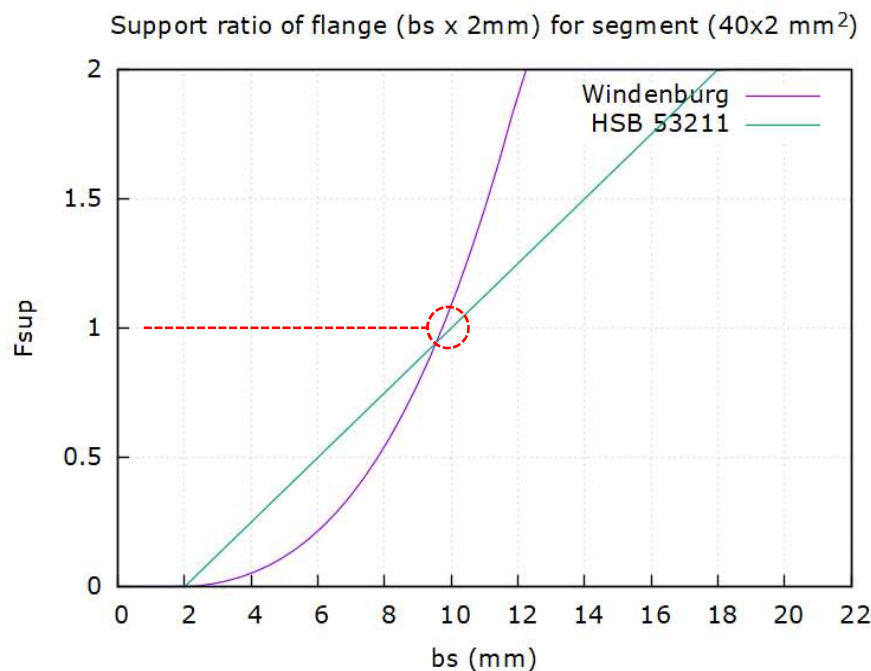


Figure 5. Support ratio : Windenburg $[2.73 \cdot I_s / (b \cdot t^3) - A_s / (b \cdot t)] / 5$ vs HSB $(b_s - t) / (0.2 \cdot a)$

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If the element is not effectively supported at any of its sides, the bi-supported element shall be split in two halves, and then treated conservatively as two separated flanges (one-edge-free). Discussion in following bullet, about flanges which are non-effectively supported, explains how to treat each of these halves.

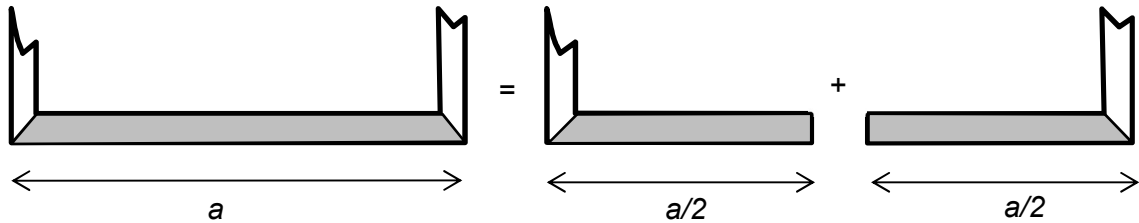


Figure 6. Bi-supported element is split in two halves if any of its sides is not effectively supported

✓ Elements supported on one side (the other is free)

For section elements with one of its sides free (flanges), the reference HSB norm does not explicitly demand for a check of lateral support. However it is considered necessary to perform this check. The method that shall be used (either if the adjacent element intending to provide support is another flange or a bi-supported element) shall be again the *Windenburg* check presented in previous sub-chapter (see §2, and Figure 3).

If the flange is not effectively supported, the reduced effectively-supported-width shall be calculated ($b_{eff-sup}$ expression in previous sub-section), and then, the total crippling load force calculated for that reduced element, will be assigned to the whole element.

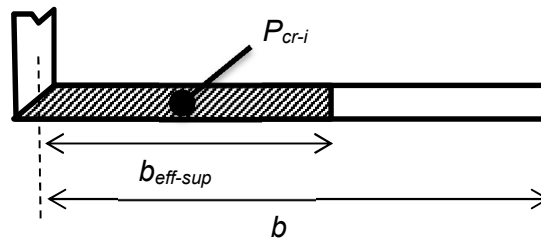


Figure 7. Flange width vs flange-effectively-supported-width, relevant for crippling analysis

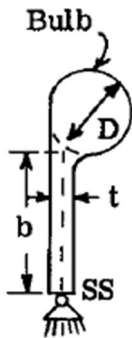
Lips

Reinforcement lips and the edge of “free” lateral sides should be treated as flanges. Effective support of main element onto the lip/flange element (and vice-versa) shall be checked before calculating final crippling strengths. The support condition of the main element will be changed to bi-supported if lip provides effective support.

Bulbs

The contribution of a circular bulb at the edge of free side, to the crippling strength, shall be done according to method in Bruhn §C.7.9.

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Bulb shall provide effective support if:

$$(D/t)^4 - 1.6 \cdot (D/t)^3 - 0.374 \cdot (D/t)^2 \geq 7.44 \cdot b/t$$

Area of bulb:

$$A_b = \pi \cdot D^2/4$$

Total area:

$$A = b \cdot t + A_b$$

Equivalent width for crippling stress calculation:

- bulb provides support (bi-supported) b
- bulb does NOT provides support (1-edge-free) $b + D$

Figure 8. Bulb effective support per Bruhn C.7.9

The crippling stress of the element shall be calculated consistently with the result of the check (edge still free or supported):

- If bulb is effective, σ_{cr} shall be calculated as bi-supported with equivalent width b in Figure 8
- If bulb is NOT not effective, σ_{cr} shall be calculated with lateral edges as supported-free with equivalent width $b+D$ in Figure 8

The cross-section area of the bulb will be added to the element, assuming it shall sustain (at least) the same crippling stress as the element, before calculating the final crippling load.

Arcs

Former/folded and extruded stringers may include not only flat elements (and bulbs) but also curved elements. For sections with circular arc-elements of one semi-circumference, crippling stresses may be calculated using actual formulation of flat elements for folded/formed profiles as follows:

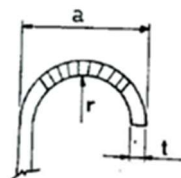
✓ 180° arc supported on both sides (bi-supported)

... using the standard formulation of the reference HSB for bi-supported elements of folded profiles when the external width decreased up to $a = 2 \cdot (r + t)$, thus:

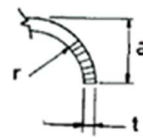
$$b = a - (t + 0.5 \cdot r) = 2 \cdot (r + t) - (t + 0.5 \cdot r) = 1.5 \cdot r - t$$

✓ 180° arc supported on one side and free on the other side

... decomposing the element in a bi-supported arc (as in previous paragraph) and a folded lip,



arc sub-element supported on both sides



arc sub-element supported on one side

Figure 9. Decomposition of circular arc element with one side free

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the equivalent width for crippling calculation of the lip is half the previous one:

$$b' = a' - 0.5 \cdot (t + 0.5 \cdot r) = (r + t) - 0.5 \cdot (t + 0.5 \cdot r) = 0.75 \cdot r - 0.5 \cdot t$$

The lip is effective support for the bi-supported arc as:

$$a' - t > 0.2 \cdot a \quad \rightarrow \quad r + t > 0.2 \cdot 2 \cdot (r + t) = 0.4 \cdot (r + t)$$

Then, the general methodology for folded/formed profile can be applied, but the following applies:

- For bi-supported elements it must be checked that $a \geq 2 \cdot (r + t)$
- For flanges (one edge free) marked as arcs, it must be true that $a = 2 \cdot (r + t)$, then the flange is decomposed in two elements (one is bi-supported, the other is a lip) as discussed above

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4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Outside width of section element (flange)
A	mm ²	Area of element or of whole section, from context
b	mm / -	Width relevant for buckling/crippling analysis of section element (web or flange) or, as subscript, refers to the bulb
$b_{eff-sup}$	mm	Reduced width of flange which is effectively supported
σ_{cr}	MPa	Crippling stress
I	mm ⁴	Bending moment of inertia
i	-	As subscript, refers to the identification number of a section element
L	-	As subscript, identifier of a lip-element intending to provide effective support to another element
P_{cr}	N	Crippling load
r	mm	Radius (typ. bend/machining internal radius of the intersection between flanges)
t	mm	Thickness of section element (flange)

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	HSB 53211-01	I	Crippling strength of short metallic sections in compression
Ref. 2	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 3	HSB 45111-01	B	Buckling of rectangular plates under various loading types and support conditions
Ref. 4	ESDU 83044	2014	Plasticity correction factors for plate buckling
Ref. 5	ESDU 76016	2013	Generalisation of smooth continuous stress-strain curves for metallic materials
Ref. 6	MANUAL DE DISEÑO CASA	01/95	“CRIPPLING” (Vol. 8 – Chapter 172)
Ref. 7	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING

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SOFTWARE

Program	Version	Description

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 30/04/2018	G. Moreda G. Baños	First issue	All pages
1.1 29/09/2021	G. Baños	Treatment of section elements which are not effectively supported is modified	Pages 3-5, 6, 8

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 08/09/2021	Florian Glock 29/09/2021

Table 3. *Approval signatures table for last revision of this chapter*